Algorithm Theoretical Basis Document (ATBD) Morphological Image Processing Approach (MIPA) for Atmospheric Boundary Layer Height (ABLH) Estimation from LiDAR/Ceilometer Data

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1 Introduction

This document describes the Morphological Image Processing Approach (MIPA) algorithm, developed for automatic Atmospheric Boundary Layer Height (ABLH) estimation from LiDAR/-ceilometer data.

This document refers to MIPA software release v1.0.0..

The key feature of MIPA is its ability to exploit the inherent spatiotemporal structure of two-dimensional LiDAR/ceilometer measurements. Rather than treating each profile independently, the algorithm leverages the continuity and topology of the LiDAR/ceilometer time series, thereby exploiting the valuable information contained in the data spatiotemporal structure.

2 Theoretical Background

2.1 Atmospheric Boundary Layer

The Atmospheric Boundary Layer (ABL), also referred to as the Planetary Boundary Layer (PBL), is the lowest portion of the troposphere that is directly influenced by the presence of the Earth's surface. Within this layer, the atmosphere responds to surface forcing, such as heating, cooling, friction, and moisture exchange, on time scales of approximately one hour or less. Consequently, the ABL exhibits pronounced temporal variability, with its depth and internal structure continuously evolving in response to meteorological conditions.

During daytime, solar heating generates buoyancy-driven turbulence that promotes vertical mixing of heat, moisture, momentum, and atmospheric constituents. Under these conditions, the ABL typically grows, reaching heights ranging from a few hundred meters to several kilometers depending on surface characteristics and atmospheric stability. Conversely, after sunset, surface cooling suppresses turbulent mixing, leading to the formation of a shallow and stable boundary layer, often only a few tens or hundreds of meters thick, while a residual layer containing aerosols and other atmospheric tracers may persist above it.

The Atmospheric Boundary Layer plays a fundamental role in numerous atmospheric processes. It governs the dispersion of pollutants, regulates the exchange of energy and mass between the Earth's surface and the free troposphere, and strongly influences weather forecasting, numerical atmospheric modeling, and air-quality assessment. Consequently, the Atmospheric Boundary Layer Height (ABLH), which identifies the upper boundary of this layer, represents a key geophysical parameter for both meteorological and environmental applications.

2.2 Aerosols as Tracers of the Atmospheric Boundary Layer

The retrieval of the Atmospheric Boundary Layer Height (ABLH) from active remote sensing observations relies on the fact that atmospheric aerosols behave, under most meteorological conditions, as effective tracers of boundary layer dynamics. Aerosol particles are continuously produced within the boundary layer from both natural and anthropogenic activities and are transported by turbulent motions. Consequently, aerosol concentration is generally contained inside the ABL and less in the free troposphere above.

As a result, the vertical distribution of atmospheric aerosols typically exhibits a relatively sharp transition around the top of the boundary layer. This transition produces a corresponding reduction of the atmospheric backscatter coefficient, which can be detected by LiDAR and ceilometer measurements. Therefore, the height at which this sharp transition occurs is usually used as a proxy for the ABLH.

The effectiveness of this approach depends on the degree of interaction between the thermodynamic structure of the atmosphere and the aerosol concentration. Under daytime hours, in most of the conditions, aerosol-based retrievals generally provide reliable estimates of the ABLH. However, during specific atmospheric situations, such as the evening transition, night hours, residual layer formation, precipitation, or weak turbulent mixing, aerosol tracers may become partially decoupled from the thermodynamic boundary layer. In these cases, the altitude identified from aerosol backscatter may differ from the boundary layer height estimated from thermodynamic variables, such as potential temperature profiles derived from radiosoundings.

Despite these limitations, aerosol backscatter measurements remain among the most widely adopted approaches for continuous ABLH monitoring. The main reason is that, unlike radiosoundings, which provide only sparse temporal sampling, LiDAR and ceilometer observations offer continuous, high-resolution measurements that enable the evolution of the boundary layer to be monitored throughout the entire diurnal cycle.

2.3 LiDAR and Ceilometer Measurements

LiDARs and ceilometers are active optical remote sensing instruments that probe the atmosphere by transmitting short laser pulses and measuring the fraction of the emitted radiation that is elastically backscattered by atmospheric molecules and particles. The received signal provides vertically resolved information on the optical properties of the atmosphere and, consequently, on the vertical distribution of aerosols and clouds.

A ceilometer is a single-wavelength elastic backscatter LiDAR operating in the near-infrared spectral region (typically between 900 and 1100 nm), characterized by low pulse energy to ensure eye-safe operation and pulse repetition frequencies in the kHz range. Compared with

high-performance LiDAR (HPL) systems, ceilometers generally provide lower radiometric performance but require significantly lower acquisition, operation, and maintenance costs. Their robustness and autonomous operation have enabled the deployment of dense observational networks, making them particularly suitable for continuous monitoring of the atmospheric boundary layer.

The quantity measured by both LiDARs and ceilometers is the atmospheric backscattered power, whose dependence on the atmospheric optical properties is described by the elastic LiDAR equation:

$$\begin{aligned} P(\lambda, z) &= \frac{K}{z^2} \cdot \beta(\lambda, z) \cdot T(\lambda, z) + P_{bkg}(\lambda) \\ &= \frac{K}{z^2} \cdot [\beta_m(\lambda, z) + \beta_p(\lambda, z)] \cdot T_m(\lambda, z) \cdot T_p(\lambda, z) + P_{bkg}(\lambda) \end{aligned} \tag{1}$$

where:

- K: calibration constant, comprising all system parameters that influence the detected backscattered signal.
- $\beta(\lambda, z)$: atmospheric backscatter coefficient.
- $T(\lambda, z)$: two-way atmospheric transmissivity.
- $P_{bkg}(\lambda)$: background optical power.

The term $P_{bkg}(\lambda)$ represents the atmospheric background signal, i.e., the sky radiance that is independent of range. It is commonly estimated as the mean signal measured at sufficiently large distances, where the atmospheric backscatter contribution is assumed to be negligible.

The atmospheric backscatter coefficient and the atmospheric transmissivity can be decomposed into their molecular and particle contributions (denoted by the subscripts m and p, respectively):

$$\begin{aligned} \beta(\lambda, z) &= \beta_m(\lambda, z) + \beta_p(\lambda, z), \\ T(\lambda, z) &= T_m(\lambda, z)\, T_p(\lambda, z). \end{aligned} \tag{2}$$

After subtracting the background contribution and compensating for the geometric spreading through multiplication by the square of the range, the attenuated backscatter signal (ABS) is obtained as

$$\begin{aligned} ABS(\lambda, z) &= \frac{\big(P(\lambda, z) - P_{bkg}(\lambda)\big) z^2}{K} \\ &= \frac{P_{rcs}(\lambda, z)}{K}, \end{aligned} \tag{3}$$

where $P_{rcs}(\lambda, z)$ is commonly referred to as the *range-corrected signal* (RCS). While the ABS is generally preferred for the quantitative retrieval of aerosol optical properties, both the RCS and the ABS preserve the same spatial distribution of the atmospheric backscatter field.

Since MIPA exploits the spatial organization of the backscatter field rather than its absolute radiometric values, the algorithm does not require calibrated observations and can therefore operate directly on the RCS. This characteristic simplifies the processing chain and enables the application of MIPA to measurements acquired by different LiDAR and ceilometer systems, even when only range-corrected signals are available.

2.4 Spatiotemporal Representation of Backscatter Measurements

LiDAR and ceilometer observations are typically acquired as continuous sequences of vertically resolved backscatter profiles. By concatenating successive profiles along the temporal dimension, the measurements can be naturally represented as a two-dimensional image, where the

vertical axis corresponds to altitude, the horizontal axis represents time, and the pixel intensity is proportional to the measured atmospheric backscatter.

In this representation, the evolution of the atmospheric boundary layer appears as a continuous interface separating regions characterized by different aerosol concentrations. Rather than being interpreted as a collection of independent one-dimensional profiles, the measurements can therefore be analyzed as a single spatiotemporal object whose structures exhibit strong spatial and temporal coherence.

This representation provides several advantages. First, the atmospheric boundary layer generally evolves continuously over time, with abrupt variations occurring only under particular meteorological conditions. Second, atmospheric structures typically extend over many adjacent profiles, making them more robust to random measurement noise than isolated one-dimensional observations. Finally, many image processing techniques are specifically designed to exploit the geometric and topological properties of two-dimensional structures, allowing the detection of coherent atmospheric features while suppressing isolated artifacts.

The MIPA algorithm is entirely based on this spatiotemporal representation. Instead of processing each profile independently, it analyzes the complete time–height image by exploiting the spatial continuity of atmospheric structures through morphological image processing and object-based analysis techniques.

Figure 1 shows an example of the input remote sensing signal arranged as a time-height image. In this representation, each column corresponds to a single LiDAR profile acquired at a given acquisition time, while each row corresponds to a fixed altitude. The image representation constitutes the input of the MIPA algorithm and allows the application of image-processing techniques to the retrieval of the atmospheric boundary layer height.

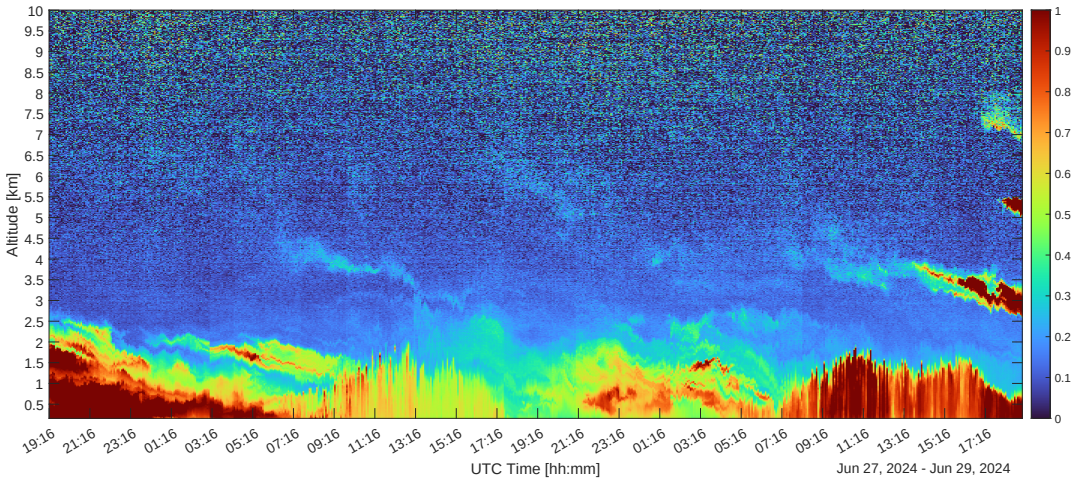


Figure 1: Example of a time–height representation of the LiDAR backscatter signal. Consecutive vertical profiles are arranged along the temporal dimension, while the signal intensity represents the measured atmospheric backscatter.

3 Algorithm Overview

The Morphological Image Processing Approach (MIPA) for ABLH estimation from LiDAR data, which was introduced in [1], validated in [2], and extended to ceilometer data in [3], consists of the following steps:

1. **Loading Data and Configuration**
   The input is a NetCDF file containing the mandatory fields described in Section 3. Users are encouraged to follow the same data format to fully exploit the algorithm. All configuration parameters are set in `config_MIPA.m`.

2. **MIPA Processing**

The steps of MIPA are summarized in the flowchart:

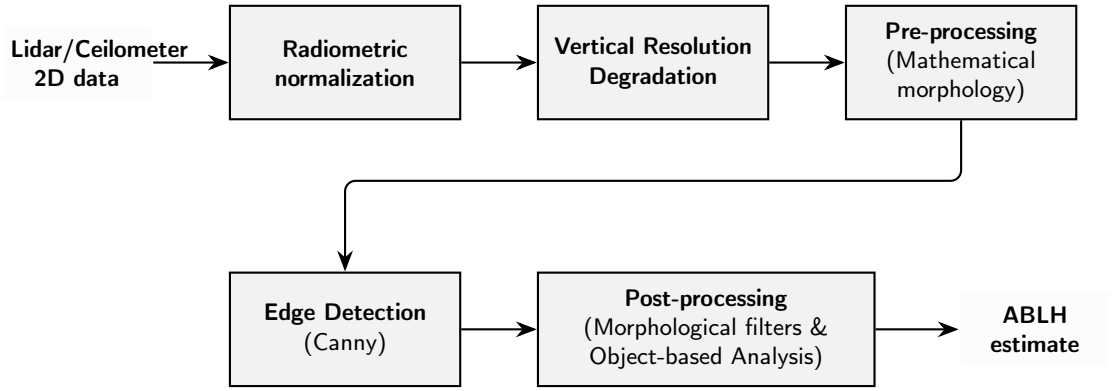
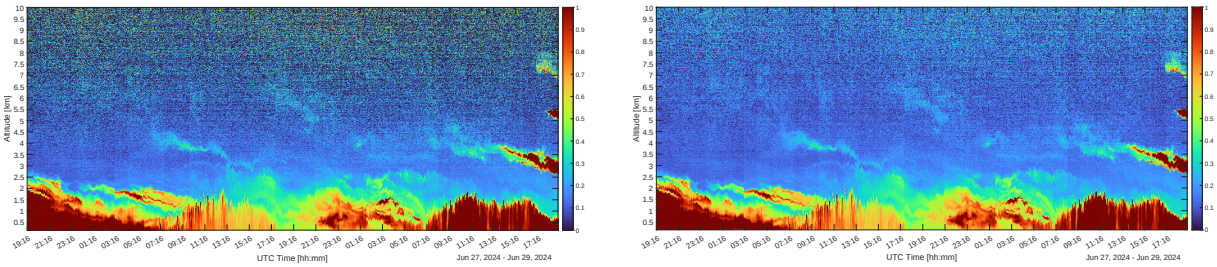


Figure 2: Processing chain of the MIPA algorithm.

- **Radiometric Normalization**
  To facilitate the subsequent processing steps, the backscatter signal is clipped using an upper-percentile threshold, with the aim of reducing the dynamic range while preserving the atmospheric structures of interest. The clipping operation is applied only within the altitude interval defined by the parameters `start_analysis_meters` and `stop_analysis_meters`, which specify the lower and upper limits of the profile considered during processing. The lower limit is typically selected to exclude the lowest portion of the atmospheric profile, where measurements may be affected by incomplete overlap or other near-range instrumental effects. After clipping the signal between the selected lower and upper percentiles and applying linear normalization, the atmospheric structures become more distinguishable while preserving their spatial organization. An example of the normalized signal is shown in Figure 3a.
- **Vertical Spatial Resolution Reduction**
  The vertical resolution is degraded to a desired lower resolution by applying a moving-average filter along each column (vertical dimension), using r as kernel support and decimation factor. Figure 3b illustrates the effect of the vertical spatial reduction. Small-scale fluctuations are attenuated while the main aerosol layers and boundary layer structures are preserved.
- **Pre-processing**
  Low-pass filtering along the time dimension (horizontal) using half-gradient filters to reduce noise while preserving vertical edges. Line structuring element of length l_{pre} is used. The result of the morphological pre-processing is reported in Figure 4. The operation reduces small isolated structures while enhancing the continuity of the atmospheric layers that are subsequently analyzed by the edge detector.
- **Edge Detection**
  The edge detection step uses, by default, the MATLAB Canny detector. Alternative implementations based on the Sobel or Prewitt operators are also possible. The binary edge map produced by the selected edge detector, is shown in Figure 5a. The detected edges identify the strongest vertical transitions in the backscatter field and constitute the initial set of candidate ABLH detections.
- **Post-processing**
  A two-step post-processing approach refines the ABLH estimation:
  - **Morphological Filtering:** Directional low-pass filters (opening + closing) remove unrealistic, rapidly fluctuating edges using line structuring elements of



(a) Radiometric normalization.

(b) Vertical resolution reduction.

Figure 3: Effect of the first pre-processing stages applied to the input backscatter image.

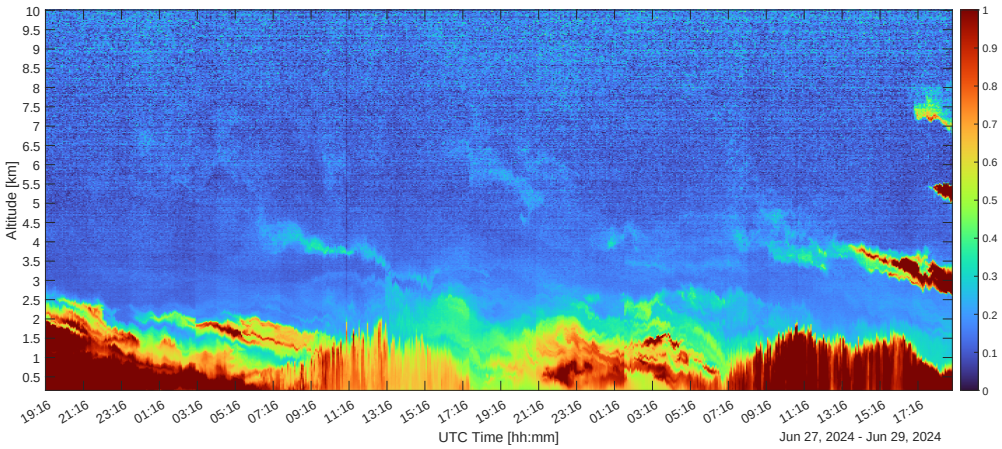


Figure 4: Backscatter image after morphological pre-processing along the temporal direction. High-frequency temporal fluctuations are attenuated while vertically oriented transitions are preserved.

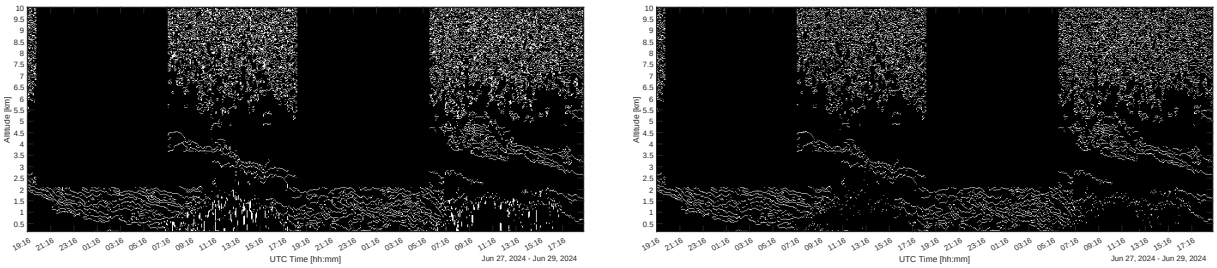
length l_{post} and angles $\theta \in [\theta_{min}, \theta_{max}]$. Figure 5b shows the effect of the post-processing stage. Isolated detections and discontinuous structures are removed, while spatially coherent edge segments are preserved for the subsequent object-based analysis.

– **Object-based Analysis:** Edge pixels connected through an 8-neighborhood are grouped into objects. Each object is characterized by its mean height, which is compared with the mean heights of the neighboring objects. Objects whose height difference exceeds the threshold

$$\delta_r = \text{round}\left(\frac{\delta_0}{r}\right)$$

are removed from the set of candidate ABLH detections. The remaining objects are finally interpolated to obtain the final ABLH estimation. Figure 6 summarizes the object-based analysis. Connected edge pixels are first grouped into individual objects according to 8-connectivity. Subsequently, each object is validated on the basis of its temporal extension and vertical consistency with neighboring objects. Objects that do not satisfy the validation criteria are discarded before the final ABLH estimation.

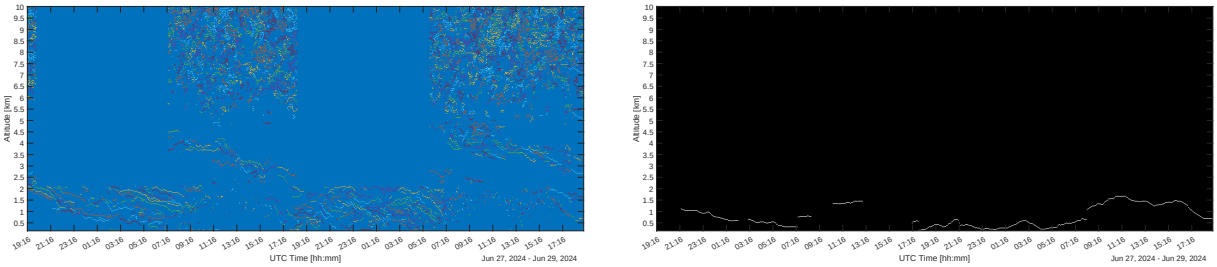
- **ABLH Estimation**
  The final ABLH estimate is obtained by interpolating the remaining validated objects. Missing values, including those occurring at the beginning and the end of the time series, are assigned the NetCDF `_FillValue`. The final output of the algorithm is



(a) Initial edge map.

(b) Post-processed edge map.

Figure 5: Effect of directional morphological post-processing on the detected edge map.



(a) Connected edge objects.

(b) Objects retained after validation.

Figure 6: Object-based analysis applied to the edge map. Connected edge pixels are grouped into objects, which are subsequently validated according to their spatial and temporal consistency.

illustrated in Figure 7, where the retrieved ABLH is superimposed on the normalized backscatter image together with the available reference observations.

3. **Saving Results**
   The output is a copy of the original NetCDF file enriched with the variable `ablh`, containing the estimated ABLH expressed as altitude above mean sea level and preserving all the metadata of the original product. The original input signal, independently of its original variable name in the input NetCDF file, is copied into the output variable `input_dataset` using the dimensions `(time, altitude)`.

   The file is saved in `data/output/` with the name `Output_<original_filename>.nc`.

4 Configuration Parameters

The behavior of MIPA is controlled through a set of configuration parameters defined in `config_MIPA.m`. Their meaning is summarized in Table 2.

5 Input and Output Specifications

In this section, the input and output data formats are described.

5.1 Inputs

The input of MIPA is a NetCDF file containing the mandatory variables listed in Table 3. Additional variables and metadata may also be present and are ignored by the processing chain unless explicitly used by the selected configuration.

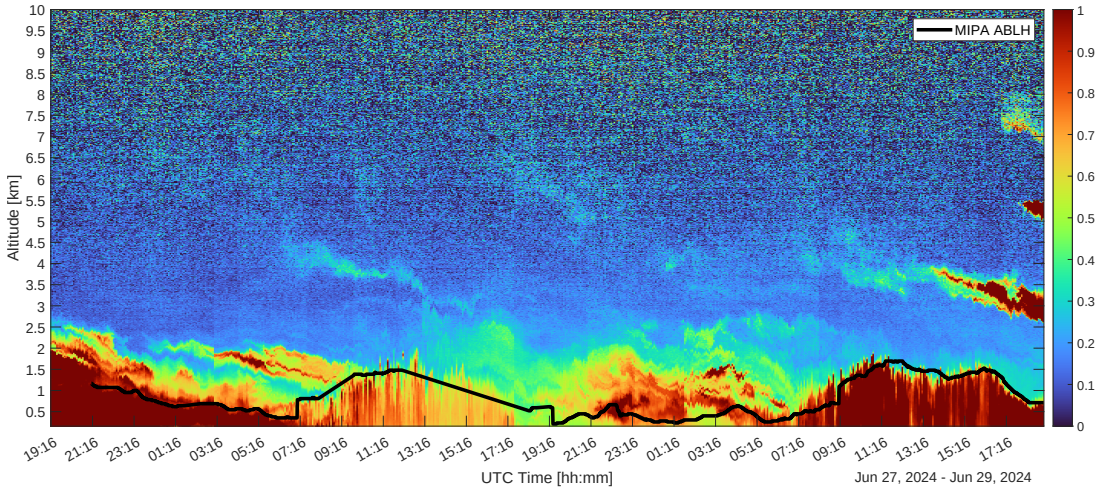


Figure 7: Final ABLH retrieved by MIPA and superimposed on the input backscatter signal. The reference ABLH values, when available, are also reported for comparison.

Variable	Dimensions	Description
Input signal	$[N_t \times N_h]$	Input remote sensing signal (variable specified by `cfg.variable_filename`)
time	$[N_t \times 1]$	Time vector
altitude	$[N_h \times 1]$	Altitude above sea level
wavelength	$[1 \times 1]$	Detection wavelength [nm]
latitude	$[1 \times 1]$	Measurement site latitude
longitude	$[1 \times 1]$	Measurement site longitude
surface_altitude	$[1 \times 1]$	Measurement site altitude above sea level

Table 3: Required input variables for MIPA inside the NetCDF file.

where N_h is the number of altitude bins and N_t is the number of time samples.

Important: The configuration parameters are defined in `config_MIPA.m`. Before running the algorithm, users must specify the measurement type through `cfg.measurement_tag`, the name of the NetCDF variable containing the input signal through `cfg.variable_filename`, and configure the processing parameters according to the characteristics of the considered dataset and instrument.

The example reported below represents only one possible implementation based on a LiDAR dataset.

```
netcdf demo_lidar {
dimensions:
    time = 2596 ;
    altitude = 5280 ;
    nv = 2 ;
variables:
    double time(time) ;
        time:units = "seconds since 1970-01-01T00:00:00Z" ;
        time:long_name = "time" ;
        time:calendar = "standard" ;
        time:axis = "T" ;
        time:bounds = "time_bounds" ;
```

Table 2: Main configuration parameters of the MIPA algorithm.

Parameter	Description	Unit
p_low	Lower clipping percentile	%
p_high	Upper clipping percentile	%
l_pre	Length of the pre-processing structuring element	down-sampled bins
l_post	Length of the post-processing structuring element	down-sampled bins
theta_min	Minimum inclination angle of the post-processing structuring element	degrees
theta_max	Maximum inclination angle of the post-processing structuring element	degrees
delta_0	Euclidean distance threshold used during object-based analysis	raw range bins
start_analysis_meters	Lower altitude limit of the processing interval	m
stop_analysis_meters	Upper altitude limit of the processing interval	m

```
    double time_bounds(nv, time) ;
    double abl_reference(time) ;
        abl_reference:units = "m" ;
        abl_reference:long_name = "planetary boundary layer altitude" ;
        abl_reference:_FillValue = 9.96920996838687e+36 ;
    double altitude(altitude) ;
        altitude:units = "m" ;
        altitude:standard_name = "altitude" ;
        altitude:long_name = "altitude above sea level" ;
    double wavelength ;
        wavelength:units = "nm" ;
        wavelength:long_name = "detection data wavelength" ;
        wavelength:_FillValue = 9.96920996838687e+36 ;
    double attenuated_backscatter(time, altitude) ;
        attenuated_backscatter:units = "m-1 sr-1" ;
        attenuated_backscatter:long_name = "total attenuated backscatter" ;
        attenuated_backscatter:coordinates = "time altitude" ;
        attenuated_backscatter:_FillValue = 9.96920996838687e+36 ;
}
```

Listing 1: NetCDF input structure.

5.2 Outputs

The output product is obtained by copying the original NetCDF file and enriching it with the MIPA retrieval together with the processing metadata. Consequently, all the variables, dimensions, and metadata originally available in the input product are preserved.

Regardless of the original name of the input signal (specified through `cfg.variable_filename`), the measurement is stored in the standardized variable `input_dataset`, whose dimensions are `(time, altitude)`. The variables added by MIPA to the output NetCDF product are summarized in Table 4.

Variable	Dimensions	Description
ablh	$[N_t \times 1]$	Retrieved Atmospheric Boundary Layer Height (ABLH) above mean sea leve
input_dataset	$[N_t \times N_h]$	Standardized copy of the original input signal
mipa_*	scalar	Configuration parameters adopted during the MIPA processing

Table 4: Variables added by MIPA to the output NetCDF product.

The retrieved ABLH is expressed as altitude above mean sea level and is therefore directly comparable with the altitude coordinate provided in the input file.

The output file is saved in `data/output/` using the name `Output_<original_filename>.nc`.

Optional figures illustrating the different processing stages and the final ABLH retrieval can be generated by setting `cfg.figures_disp_flag = "true"` in `config_MIPA.m`. Setting the parameter to `"false"` disables the generation of the figures.

```
netcdf Output_demo_lidar {

variables:

    double input_dataset(time, altitude) ;
        input_dataset:units = "1" ;
        input_dataset:long_name =
            "input remote sensing dataset in arbitrary units" ;
        input_dataset:coordinates = "time altitude" ;
        input_dataset:_FillValue = 9.96920996838687e+36 ;

    double ablh(time) ;
        ablh:units = "m" ;
        ablh:long_name =
            "atmospheric boundary layer altitude" ;
        ablh:_FillValue = 9.96920996838687e+36 ;

    int mipa_upper_percentile ;
        mipa_upper_percentile:long_name =
            "applied percentile upper limit for clipping" ;
        mipa_upper_percentile:units = "1" ;

    int mipa_lower_percentile ;
        mipa_lower_percentile:long_name =
            "applied percentile lower limit for clipping" ;
        mipa_lower_percentile:units = "1" ;

    int mipa_down_sampling_factor ;
        mipa_down_sampling_factor:long_name =
            "applied image down-sampling factor" ;
        mipa_down_sampling_factor:units = "1" ;

    int mipa_pre_processing_structuring_element_length_factor ;
        mipa_pre_processing_structuring_element_length_factor:long_name =
            "applied pre-processing structuring element length factor in units of
    down-sampled resolution" ;
        mipa_pre_processing_structuring_element_length_factor:units = "1" ;

    int mipa_post_processing_structuring_element_length_factor ;
        mipa_post_processing_structuring_element_length_factor:long_name =
            "applied post-processing structuring element length factor in units of
    down-sampled resolution" ;
        mipa_post_processing_structuring_element_length_factor:units = "1" ;

    int mipa_post_processing_structuring_element_minimum_angle ;
```

```
        mipa_post_processing_structuring_element_minimum_angle:long_name =
            "applied post-processing structuring element minimum angle with respect
    to horizontal direction" ;
        mipa_post_processing_structuring_element_minimum_angle:units = "degrees" ;

    int mipa_post_processing_structuring_element_maximum_angle ;
        mipa_post_processing_structuring_element_maximum_angle:long_name =
            "applied post-processing structuring element maximum angle with respect
    to horizontal direction" ;
        mipa_post_processing_structuring_element_maximum_angle:units = "degrees" ;

    int mipa_euclidean_distance_threshold ;
        mipa_euclidean_distance_threshold:long_name =
            "applied euclidean distance threshold among object and related
    neighborhood objects in units of raw resolution" ;
        mipa_euclidean_distance_threshold:units = "1" ;

}
```

Listing 2: Variables added or modified by MIPA in the output NetCDF product.

6 Demonstration

The MIPA pipeline can be tested using the demonstration datasets provided with the software package. Two standardized NetCDF products are included: one for a LiDAR system and one for a CHM15k ceilometer. Both demonstration files contain the standardized variable `input_dataset`, which stores the input signal required by the algorithm.

6.1 Running the Demo

1. Navigate to the `src` folder and run the main script:

   ```
   main_MIPA.m
   ```

2. The demonstration NetCDF files are provided in `data/input/`:

   ```
   demo_lidar.nc
   demo_chm15k.nc
   ```

3. In the configuration file `config_MIPA.m`, ensure the following parameters are set:

   ```
   cfg.filename = 'demo_lidar';
   cfg.measurement_tag = 'lidar';
   cfg.variable_filename = 'input_dataset';
   ```

   Note: The parameter `cfg.filename` must match the selected input file, `cfg.measurement_tag` must identify the measurement type, and `cfg.variable_filename` must correspond to the NetCDF variable containing the input signal. The same procedure can also be applied to the CHM15k demonstration dataset by selecting the appropriate input file and setting `cfg.measurement_tag = 'ceilometer'`.

4. After processing, the output is saved in `data/output/` as

   ```
   Output_<input_filename>.nc
   ```

6.2 Example Output

The output NetCDF file contains the retrieved atmospheric boundary layer height in the variable `ablh`, together with the standardized input signal (`input_dataset`) and the processing parameters described in Section 4. An example of the final ABLH retrieval is shown in Figure 7.

7 Repository

This document describes MIPA software release v1.0.0. The latest version of the source code, documentation, and example data is available from the GitHub repository:

`https://github.com/username/mipa-repository`

8 Software Requirements

The software has been tested and verified on MATLAB R2024b (version 24.2).

- **MATLAB** $\geq$ version 24.2 (R2024b)
- **Image Processing Toolbox** $\geq$ version 24.2

Users with earlier versions of MATLAB or the required toolboxes may experience compatibility issues.

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If you use this software in scientific work, please cite the following publications:

- For LiDAR data: **[1]** or **[2]**
- For ceilometer data: **[3]**.

10 References

[1] G. Vivone, G. D'Amico, D. Summa, S. Lolli, A. Amodeo, D. Bortoli, and G. Pappalardo. Atmospheric boundary layer height estimation from aerosol lidar: a new approach based on morphological image processing techniques. *Atmospheric Chemistry and Physics*, 21(6):4249–4265, 2021.

[2] Giuseppe D'Amico, Alberto Arienzo, Gemine Vivone, Aldo Amodeo, Francesco Cardellicchio, Pilar Gumà-Claramunt, Benedetto De Rosa, Paolo Di Girolamo, Ilaria Gandolfi, Aldo Giunta, Teresa Laurita, Fabrizio Marra, Lucia Mona, Michail Mytilinaios, Nikolaos Papagiannopoulos, Marco Rosoldi, and Donato Summa. Atmospheric boundary layer height estimation from lidar observations: Assessment and validation of mipa algorithm. *Remote Sensing*, 17(16), 2025.

[3]